What Does Public-Data ESG Measure? A Transparent Multi-Factor Mid-Cap Index, Its In-Sample Validation Pitfalls, and an Out-of-Sample Test

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***Abstract*—Open research pipelines increasingly build ESG scores for mid-cap firms from public data. We ask what such scores measure and how they should be validated. We build a reproducible nine-factor ESG–financial index for 269 US and Indian mid-caps with cell-level provenance, and audit it.**

Construct validity is weak. **Only 21 of 34 ESG indicators carry information. Most E and S cells are financial proxies, and the other factors explain $R^2 = 0.42$ and 0.51 of those pillars.**

In-sample validation misleads. **Trailing-return "ICs", a contemporaneous "forward" proxy and a misapplied bootstrap made the index look strong.**

Out of sample, the index does not predict returns. **In a hold-out window (2026-04-02 to 2026-09-23, $N = 260$) with hypotheses fixed in advance, the composite's IC is -0.094 (95% CI $[-0.219, 0.036]$). The IC is negative under 96.7% of random weightings.**

ESG predicts lower volatility, but not as ESG. **Higher ESG predicts lower volatility beyond trailing volatility (partial $\rho = -0.163$, Holm $p = 0.013$), even with 8 risk controls and sector effects. The signal comes from financial-proxy cells, not measured ESG data (partial $\rho = 0.032$). It adds only 0.0056 to out-of-fold R^2 and weakens given recent volatility. Meanwhile, an ESG-only top-19 portfolio trailed the universe by 19.9 percentage points.**

Public-data ESG thus acts as a slow-moving low-risk tilt, not as ESG information or alpha. We pre-register a confirmatory second window, and one command regenerates every number in the paper.

***Index Terms*—ESG measurement, construct validity, composite indicators, out-of-sample validation, pre-registration, mid-cap equities, reproducibility**

I. Introduction

Sustainable-investment assets reached about $30 trillion in the markets covered by the most recent global review [1]. Yet the ESG ratings that steer this capital disagree strongly across providers: the average pairwise correlation is about 0.54 [2], [3]. Ratings are also proprietary, so their construction cannot be inspected [4], [5]. The gap is widest for mid-caps. Large caps increasingly face mandatory sustainability reporting, while mid-caps are covered by fewer analysts and data vendors, even though they are large enough to be investable.

A natural response, and a growing practice in open-source quantitative research, is to build ESG scores from public data. Such pipelines combine regulatory filings, free governance scores and financial-statement ratios with documented economic rationales, such as "higher operating margin within a sector implies lower emissions intensity" [6]–[8]. These pipelines are transparent, but transparency is not validity. Two questions are rarely asked of them:

1) What does the resulting "ESG" score measure, once each input's actual source is traced?
2) Is the evidence usually offered for such indices valid, i.e. can it tell a good composite from a bad one?

We study both questions on a concrete, fully reproducible system: a nine-factor ESG–financial composite index for 269 mid-cap equities in the US and India, scored on a data snapshot taken at the close of 2026-04-02. The index combines SASB-weighted ESG pillars [9], [10] with eight financial factors and three investor profiles. It follows composite-indicator practice [11]: robust normalisation, explicit weights and provenance tracking. Every data cell carries a source label, and every number in this paper is generated from pipeline outputs by a single script.

We then audited the pipeline end to end, as one would audit any published index. The audit found implementation errors that changed the scores materially. The most consequential were indicators whose direction was flipped twice, which rewarded high emitters and high governance-risk firms. It also found a family of *validation* errors: designs that report favourable numbers whether or not the index works. We fixed the implementation errors and replaced the validation evidence with a genuine out-of-sample (OOS) test. The universe, weights and three hypotheses were fixed from the snapshot before any post-snapshot price was downloaded.

Research questions. Accordingly, we ask:

RQ1 (Construct validity) What does a public-data ESG score for mid-caps measure, and how much of it is financial

information already present in the other factors?

RQ2 (Validation) How far do common in-sample validation designs overstate the quality of an ESG–financial composite?

RQ3 (Out of sample) Over the following 118 trading days, does the composite predict returns, and does its ESG component predict risk?

RQ4 (Mechanism) If ESG predicts risk, is that ESG information, a known risk factor, or past volatility carried forward?

Figure 1 shows how the study design maps onto these questions.

Contributions.

1) *An open, provenance-tracked index pipeline* for US and Indian mid-caps. It is reproducible from public sources, and its paper artifacts are generated from its outputs (Section III).
2) *A construct-validity audit of public-data ESG.* Only 21 of 34 declared indicators carry information. The E and S pillars are mostly re-expressed financial ratios (R^2 on the other factors of 0.42 and 0.51). Only governance, built from ISS scores (209 firms) and ownership data, is close to orthogonal to them ($R^2 = 0.03$) (Section V-A).
3) *A quantification of in-sample validation pitfalls.* These are trailing-return "returns", a contemporaneous "forward" proxy, a correction fitted to the validation target, and a misapplied bootstrap. They made a composite with no OOS predictive power look like a strong one (Section V-B).
4) *A hold-out test with hypotheses fixed in advance*, Holm-corrected inference, bootstrap intervals and survivorship bounds. The composite does not predict returns, under the deployed or almost any other weights; the ESG component predicts lower risk (Sections V-C and V-E).
5) *A decomposition of the ESG risk signal* by factor, data source and horizon. The signal survives 8 known risk characteristics and sector effects. But it lives in financial-proxy cells, not in measured ESG data, adds little to a volatility forecast, and fades given recent realised volatility (Section V-D).
6) *A pre-registered second window*, with hashed scores, plan and code, which turns the next six months into a confirmatory test (Section IV-D).

The results are negative on returns and qualified on risk. ESG as built here does predict risk, as the view that ESG operates mainly through risk suggests [12], [13]. But the part of the score that does so is financial information and slow-moving volatility, not environmental or social measurement. The results also bear out warnings about false discoveries and backtest overfitting in factor research [14], [15]. Our practical message is simple. A transparent composite can still be invalid. The evidence that tells the two apart is out-of-sample, and even then it must be traced back to its data sources before it is credited to ESG.

II. Related Work

A. ESG Rating Divergence and Measurement

Commercial ESG ratings disagree substantially. Berg, Kölbel and Rigobon [2] attribute the divergence mainly to measurement and scope differences rather than weighting. Chatterji et al. [3] document low convergence, and disagreement itself carries return and risk implications [16], [17]. Christensen, Serafeim and Sikochi [5] show that more disclosure can *increase* disagreement, and Kotsantonis and Serafeim [4] catalogue data problems such as inconsistent peer sets, imputation and scale. This literature treats ESG ratings as noisy, model-dependent measurements. We apply the same lens to a public-data ESG score, where every modelling choice is visible and can be audited cell by cell.

B. ESG, Returns and Risk

Meta-analytic evidence reports predominantly non-negative ESG–performance associations [18], [19]. Materiality matters: industry-material issues drive the effect [9]. Equilibrium models imply that investor tastes for sustainability can *lower* the expected returns of green assets while making them hedges [20], [21], and carbon exposure is priced [22]. On the risk side, CSR is associated with lower systematic risk [12], and high-trust firms fared better in the 2008–2009 crisis [13]. A risk association alone does not identify an ESG effect, however. Low-volatility and low-beta stocks [23], [24] and high-quality firms [25] have distinct risk and return profiles, and realised volatility is highly persistent. An ESG score that loads on these characteristics will "predict" risk without containing ESG information. We therefore test the risk channel against these characteristics and against recent volatility, and trace it back to the sources of the score's inputs.

C. Public-Data and Proxy ESG

Where direct disclosure is sparse, researchers infer ESG-relevant behaviour from observables. Examples include environmental performance and firm value [6], eco-efficiency [26], [27], employee satisfaction [7], entrenchment indices [28], machine-learned ESG scores [8] and text-mined sustainability reports [29]. Each proxy has a rationale, but aggregating many proxies built from the *same* financial statements risks producing an ESG score that is a relabelled financial-quality score. We quantify this risk directly (Section V-A).

D. Multi-Factor Models and Composite Indicators

Factor models explain the cross-section of returns with a small number of characteristics [30]–[34]. Commercial ESG indices tilt or exclude constituents on ESG ratings under tracking-error constraints [35]–[37]. Our index is a multi-criteria *ranking* rather than a pricing model. Its construction follows the OECD handbook for composite indicators [11]: normalisation, weighting, aggregation and sensitivity analysis.

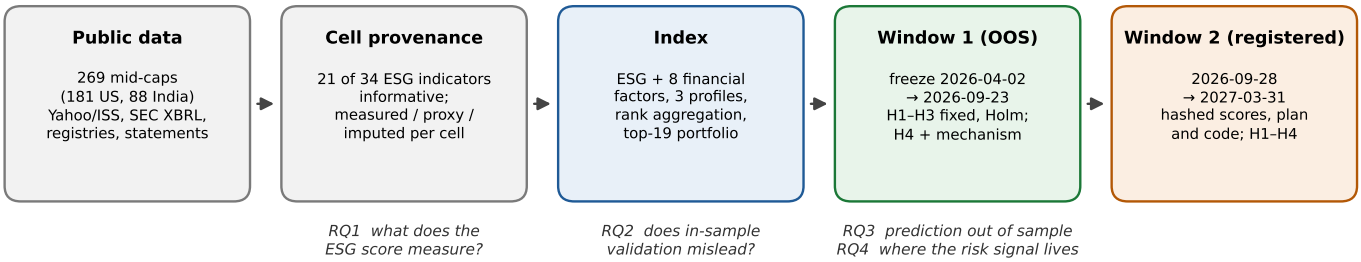


Fig. 1. Study design. Scores are frozen at the 2026-04-02 close and evaluated on the following window (RQ3–RQ4). A second window, whose scores, plan and code were hashed before it opened, provides a confirmatory test.

E. Validation, Multiple Testing and Overfitting

Hundreds of published factors fail to replicate once multiple testing is taken into account [14]. Backtests that choose among many configurations overfit with high probability [15]. The remedy proposed across empirical sciences is to fix the hypotheses and the analysis plan before outcomes are observed [38]. We adopt this remedy in two steps. The first is a hold-out window with hypotheses fixed before outcomes were computed and a family-wise error correction [39]. The second is a formally pre-registered window whose scores, plan and code are hashed before it opens. We also show concretely how in-sample designs common in index research produce favourable evidence irrespective of predictive validity.

III. Data and Index Construction

A. Universe and Snapshot

The candidate universe contains 321 firms. These are mid-cap constituents drawn from the S&P MidCap 400, the Russell Midcap and the NIFTY Midcap 150 [40], plus 45 S&P 500 large caps that serve only as a reference group in normalisation. All fundamentals and trailing market statistics were downloaded from Yahoo Finance [41], [42] and SEC EDGAR [43] and are dated at the close of 2026-04-02. We verified the date by matching every cached last price to the exchange close. We apply one investability filter: a firm must have traded on the snapshot date. This removes 7 firms (ANSS, SAMSN, CMA, CHX, ZI, WELSPUNIND, SUVENPHAR) that had already been acquired or delisted. Their presence would have let the evaluation window "see" corporate events that were already known. The scored universe has 314 firms, of which 269 are mid-caps (181 US, 88 India) in 11 sectors. All analysis below concerns the mid-caps. Indian monetary fields are converted at the 2026-04-02 close of INR/USD (92.97); ratios are unaffected.

B. Public-Data ESG with Cell-Level Provenance

The ESG schema declares 34 indicators (11 E, 11 S, 12 G). Each company–indicator cell is filled by the first available source in a fixed priority order, and the source is recorded:

1) EPA Toxics Release Inventory (US facilities).
2) SEC XBRL filing items.
3) ISS governance QualityScores redistributed by Yahoo Finance, namely audit, board, compensation, shareholder-rights and overall risk on a 1–10 scale, stored as $(10 - r) \times 10$ so that higher means better.
4) Yahoo ownership and analyst-coverage fields.
5) Binary membership of public initiatives (CDP, SBTi, RE100, UN Global Compact).
6) Financial proxies: within-sector percentiles of financial ratios, each with a published rationale.
7) Within-sector median.
8) Global median.

TABLE I
Provenance of ESG cells used in scoring (% of cells per pillar, mid-cap universe). "Used/declared" counts indicators with cross-sectional variation.

Source of cell	E	S	G
Indicators used / declared	4/11	7/11	10/12
ISS governance score	–	–	39
Yahoo ownership/coverage field	–	–	28
SEC XBRL item	–	–	3
Public registry (0/1)	25	14	–
Financial ratio	–	14	–
Financial proxy	75	70	12
Imputed	<1	1	19

No random noise is added at any tier. The mapping from each indicator to its actual source is given in Appendix A. Table I summarises the outcome for the mid-cap universe. Only 21 indicators receive cross-sectional variation. No mid-cap matched an EPA facility record, and the Sustainalytics pillar scores that Yahoo once exposed are no longer populated. Consequently the E and S pillars consist almost entirely of financial ratios and financial proxies. The G pillar's largest source is ISS scores (39% of cells, available for 209 of 269 firms), followed by ownership and analyst-coverage fields.

C. Factor Scores

Each indicator is winsorised at the 1st/99th percentile and standardised with a robust z-score, $z = (x - \mathrm{med}(x))/(1.4826\,\mathrm{MAD}(x))$, clipped to $[-3, 3]$ [44]. Lower-is-better indicators are sign-flipped only when they are stored

as raw magnitudes. Section III-E shows why this qualification matters.

ESG composite. Indicators are averaged within categories (e.g. emissions, energy, labour, board), categories are weighted into E, S and G, and the pillars are combined with sector-specific SASB materiality weights (Appendix B) [9], [10]. Before the G pillar is formed, governance values are quantile-harmonised by source so that the scales of ISS and SEC-derived inputs do not differ. The composite is re-standardised to mean 50 and standard deviation 10 over the scored firms and clipped to $[0, 100]$; because aggregation uses ranks (Eq. (1)), this scale does not affect the index.

Eight financial factors.

- *Financial*: profitability, margins, cash, payout.
- *Operational*: revenue per employee, R&D intensity, FCF and gross margin, headcount.
- *Risk-adjusted*: 1-year Sharpe and Sortino ratios, maximum drawdown, Amihud illiquidity [45], skewness, kurtosis.
- *Growth*: revenue, earnings and sustainable growth.
- *Value*: inverted P/E, P/B, P/S, EV/EBITDA, EV/revenue.
- *Stability*: liquidity, leverage, 30-day volatility.
- *Market*: liquidity, beta and volatility, and 1-, 3- and 6-month momentum.
- *Sector position*: the within-sector percentile of the average within-sector z-score of four raw size and efficiency ratios.

Market score is computed but receives zero weight in every profile. This avoids scoring firms on the same trailing returns that earlier versions of the pipeline used as the validation target.

D. Aggregation and Investor Profiles

For profile p with weights $w^{(p)}$, the preference score is a weighted mean of factor percentiles:

$$C_i^{(p)} = \sum_k w_k^{(p)} \frac{100 \ \mathrm{rank}(F_{ik})}{N}, \qquad \sum_k w_k^{(p)} = 1, \quad (1)$$

with average ranks for ties and median fill for missing factors. The ranks are taken over the $N = 314$ scored firms, so the 45 large caps act as a fixed reference group. Table II lists the effective weights of the three profiles. The balanced profile puts 23.2% on ESG. The equal-weight portfolio of the top $\lceil 7\% \cdot N \rceil = 19$ mid-caps is the index portfolio.

Weight provenance. The profile weights were set by the authors using in-sample rank correlations with *trailing* momentum, a practice we criticise in Section V-B. They were last revised in July 2026, after the snapshot. For this reason the OOS analysis also reports the February 2026 weights, which predate the snapshot, and equal weights.

E. Audit Corrections

Before any OOS number was computed, we audited the pipeline and corrected the following (full list in Appendix C):

1) **Double direction flips.** `scope1_emissions` and `esg_risk_rating` were already stored as "higher is better" scores, and were flipped again. This rewarded high-release and high-governance-risk firms. The Spearman correlation between the pre- and post-audit E pillars is -0.45; for G it is 0.38.
2) **Synthetic noise in imputation.** Median imputation added multiplicative Gaussian noise, and registry memberships were encoded as noisy scores rather than 0/1.
3) **Fitting to the validation target.** A "monotonicity correction" fitted each factor by isotonic regression to the proxy later used to validate it (Section V-B).
4) **Hidden score transforms.** An undocumented gate set the bottom 30% of firms by financial score to zero, and an exponent of 1.5 was applied to preference scores. Both were removed.
5) **Stale inputs and look-ahead.** A live, non-snapshot FX rate was used, and delisted firms were retained (Section III-A).

TABLE II
EFFECTIVE PROFILE WEIGHTS (%) AFTER REMOVING THE MARKET FACTOR.

Factor	ESG-first	Balanced	Financial-first
ESG composite	56.1	23.2	5.6
Financial	8.2	18.9	27.8
Risk-adjusted	8.2	15.8	20.0
Growth	6.1	14.7	16.7
Stability	6.1	8.4	7.8
Sector position	7.1	8.4	5.6
Operational	5.1	6.3	8.9
Value	3.1	4.2	7.8
Market	0.0	0.0	0.0

The corrected balanced composite has a rank correlation of 0.95 with the pre-audit composite, and 13 of 19 top-19 names are shared.

IV. Validation Design

A. In-Sample Diagnostics

We report conventional diagnostics on the snapshot cross-section: variance-inflation factors, PCA, and agreement between profiles. We also re-run the pipeline's former validation designs, to measure how much they inflate apparent quality. Those designs are:

1) Rank "ICs" against 1-, 3- and 6-month *trailing* returns, which end on the snapshot date.
2) ICs against a `forward_quality_proxy`. Despite its name, this is a *contemporaneous* composite of ROA, ROE, margins, growth and liquidity, inputs shared with several factors.
3) Quintile monotonicity tests against the same proxy.

B. Out-of-Sample Test

Window and outcomes. Scores are fixed at the 2026-04-02 close. Outcomes are measured over the window (2026-04-02, 2026-09-23]: 118 US and 122 Indian trading days. We use split- and dividend-adjusted closes and compute:

- returns at 1 and 3 months and over the full window;
- realised volatility, maximum drawdown, and beta against the S&P 500 or NIFTY 50.

Returns are demeaned within country, because the two markets had different index returns (17.1% and 3.2%). Sector-and-country neutralisation is reported as a robustness check. 9 firms were acquired during the window (CFLT, CWAN, EXAS, HOLX, WBS, HLX, CTRA, SEE, EA) and have no end price. For them, we report bounds that assign the 5th or 95th percentile of the return distribution.

Hypotheses. Three hypotheses were fixed before outcomes were computed. All are one-sided, with Holm correction across the family [39].

H1 The balanced composite has a positive rank IC with the country-neutral full-window return.

H2 Given trailing (2-year) volatility, higher ESG predicts lower realised OOS volatility (partial Spearman).

H3 The top-19 balanced portfolio beats the equal-weighted universe.

Confidence intervals come from 5,000 firm-level bootstrap resamples; in H3, the portfolio is re-formed in each resample. All other analyses are secondary: factor ICs by horizon (Holm-corrected within horizon), portfolios by profile, and ESG–risk partial correlations and rank regressions with sector and country fixed effects and HC3 errors.

Power. With $N = 260$, a two-sided test at $\alpha = 0.05$ has 80% power only for $|\rho| \geq 0.173$. ICs of the size usually reported for single factors (0.02–0.10 [46]) are therefore not detectable in one cross-section. A null for H1 means "no large effect", not "no effect".

Transparency about the design. The window had already ended when the hypotheses were written down, and the design is not externally pre-registered [38]. Its safeguards are procedural:

- universe, weights and scores are frozen artefacts dated at the snapshot;
- prices after the snapshot were downloaded only after the hypotheses and code were fixed;
- all alternative weightings are reported.

C. Secondary Analyses: Is the Risk Signal ESG?

If H2 holds, three alternative explanations must be ruled out before the effect can be attributed to ESG. All tests below are exploratory and were specified after H1–H3 had been evaluated.

1) *Known factors.* ESG may proxy for a known low-risk characteristic [23]–[25]. We regress the percentile rank of realised volatility on the ESG rank and on the ranks of trailing volatility, trailing beta, size, quality (profitability score), leverage (debt/equity), liquidity (log dollar volume), six-month momentum and value (price/book), with sector and country fixed effects and HC3 errors (*H4*). We also ask whether ESG improves an out-of-fold forecast of realised volatility (10-fold cross-validation, 50 repeats), and we double-sort firms on trailing volatility (terciles within country) and ESG (above or below the median within each tercile).
2) *Provenance.* The signal may come from the financial proxies rather than from measured ESG data. For each firm we average the percentile ranks of its indicator cells separately by source: *measured* (ISS, SEC, public registry; 8.9 cells per firm on average), *proxy* (financial-statement proxies; 10.1) and *imputed* (sector or global medians; 1.9). We then re-run the risk tests on each sub-score.
3) *Stale volatility.* ESG may simply be a slow-moving record of past volatility. We split the window at 2026-06-28 and ask whether ESG predicts second-half volatility given *first-half* realised volatility, which the scores never saw but an investor would have.

Firms whose realised volatility fell below a quarter of their trailing volatility (PEN, AES) were trading at announced takeover prices; we report results without them, by country, with winsorised volatility, and without utilities and real estate.

For the return null we report three checks:

- the distribution of the OOS IC over 2000 weight vectors drawn uniformly from the simplex, and over a sweep of the ESG weight from 0 to 1;
- month-by-month ICs;
- top-19 excess returns net of round-trip costs of 16 bps (US) and 50 bps (India, including securities transaction tax), and three times those costs.

D. Pre-Registered Second Window

To make the next test confirmatory, we froze the 269 firm scores and a four-hypothesis analysis plan on 2026-09-25, before the second window (2026-09-28, 2027-03-31] opens. The plan, the frozen scores and the evaluation code are hashed (SHA-256); the package is in `preregistration/window2` of the repository. H1–H3 carry over unchanged. H2 and H4 update their controls to what is observable at the anchor date, namely six months of realised volatility, beta and return. The evaluation script refuses to run before the window closes or if any registered file has changed. A dry run of the same code on (2026-07-01, 2026-09-23] is included in the registration.

E. Rank Uncertainty

Rank intervals combine two sources of uncertainty, estimated over $B = 1000$ draws:

- *sampling*: the reference universe used for percentiles is resampled with replacement;
- *weights*: profile weights are drawn from a Dirichlet(κw) with $\kappa = 100$, i.e. about $\pm 20\%$ relative variation for a 0.2 weight.

The re-implementation reproduces the deployed composite with Spearman $\rho = 0.999$.

V. Results

A. RQ1: What the ESG Score Measures

Sources. Table I shows how little of the ESG schema is populated by ESG data. The E pillar has four informative indicators:

- three within-sector financial percentiles: operating margin, standing in for emissions; revenue-to-market-value, standing in for renewable share; and gross margin, standing in for waste;
- one binary flag for public climate commitments.

The S pillar has seven indicators, six of them financial. Examples are revenue per employee (which enters twice, once as a ratio and once as a proxy for employee satisfaction), dividend-plus-FCF yield for community investment, and payout-plus-current-ratio for supply-chain audits. The G pillar differs. 39% of its cells are ISS governance risk scores, and the rest are ownership, analyst-coverage and SEC items or imputations.

Redundancy. Regressing each pillar on the other seven factor scores gives $R^2 = 0.42$ (E), 0.51 (S) and 0.03 (G); for the composite it is 0.51. The ESG composite has a Spearman correlation of 0.58 with the financial score. The pillars are nearly unrelated to each other: E–S $\rho = 0.42$, E–G $\rho = 0.01$, S–G $\rho = 0.12$. Hence "ESG" here is two different objects:

- E and S: financial quality measured within sector;
- G: largely independent third-party governance information.

Conventional diagnostics do not reveal this. The maximum VIF among factor scores is only 2.04 (ESG composite), and the first three principal components explain 64.9% of factor variance, which looks like a well-conditioned, multidimensional factor set. The redundancy sits *inside* the ESG block, where the financial proxies duplicate information already present in the financial factors.

Profiles. The three investor profiles are highly concordant. The balanced composite has Spearman correlations of 0.90 with the ESG-first and 0.96 with the financial-first composite. The top-19 lists share 14 and 14 names, respectively. Changing the ESG weight from 5.6% to 56.1% therefore changes the ranking far less than the labels suggest.

B. RQ2: In-Sample Validation Overstates Quality

Table III reports the pipeline's former validation evidence next to the OOS outcome. The balanced composite's in-sample correlation with the trailing six-month return is 0.23. This is not evidence of prediction: the "returns" end on the snapshot date, and the composite contains risk-adjusted and stability scores built from the same price history. The correlation with the contemporaneous quality proxy is 0.50. It is inflated further by shared inputs, since the proxy is built from ROA, ROE, margins and growth, which also feed the financial, growth and operational factors. Out of sample, all three profiles have negative point ICs, and every confidence interval contains zero.

Table IV isolates three further artefacts.

TABLE III
In-sample "validation" versus out-of-sample outcome. In-sample columns are rank correlations of the composite with trailing six-month returns and with a contemporaneous quality proxy on the snapshot cross-section.

	In-sample (pre-snapshot)		Out-of-sample	
Composite	vs. trailing 6m	vs. quality proxy	IC, full window	95% CI
Balanced composite	0.227	0.503	−0.094	[−0.227, 0.040]
ESG-first composite	0.093	0.272	−0.111	[−0.230, 0.015]
Financial-first composite	0.288	0.580	−0.063	[−0.186, 0.073]

TABLE IV
Diagnostics before and after the audit corrections of Section III-E. Pre-audit rank intervals come from the former "Bayesian bootstrap"; post-audit intervals from Section IV-E.

Diagnostic	Pre-audit	Post-audit
Factors monotonic in quintiles of the quality proxy	7/8	3/8
Median 95% rank-CI width, all firms	248	42
Maximum VIF among factor scores	2.80	2.04
Top-19 balanced portfolio: firms in common	–	13/19
Balanced composite: Spearman ρ pre vs. post	–	0.95
Out-of-sample IC of the balanced composite	−0.110	−0.094

1) The isotonic monotonicity correction was fitted to the quality proxy. With it, 7 of 8 factors were "monotonic" in that proxy; without it, 3 are.
2) The former "Bayesian bootstrap" multiplied each firm's score by i.i.d. Exponential(1) noise. The resulting rank intervals had a median width of 248 positions out of 276. The paper nonetheless reported perfect rank stability (Kendall $\tau = 1$) from a separate calculation.
3) The corrected composite shares only 13 of its top-19 names with the pre-audit version, although the Spearman correlation of the two is 0.95. The direction errors changed the E pillar most (pre–post $\rho = -0.45$).

None of these artefacts changed the OOS conclusion: the pre-audit composite also has no predictive power (IC $= -0.110$). They changed only the *evidence* that was offered for the composite.

C. RQ3: Out-of-Sample Evidence

Over the window, the S&P 500 returned 17.1% and the NIFTY 50 3.2%. The equal-weighted mid-cap universes returned 15.8% (US) and 18.9% (India). Table V reports the three primary hypotheses.

H1 and H3 (returns): not supported. The balanced composite's IC with the country-neutral full-window return is -0.094 (95% CI $[-0.219, 0.036]$). Its top-19 portfolio beat the universe by 1.9 percentage points (pp), with CI $[-10.9, 11.5]$. Both estimates are robust to the 9 M&A exits: under the survivorship bounds, the IC lies in $[-0.090, -0.083]$ and the excess in $[-3.4, -1.3]$ pp. The top-ranked firm (CTRA) was among the exits.

H2 (risk): supported. Controlling for trailing volatility, higher ESG predicts lower realised volatility: partial $\rho = -0.163$, 95% CI $[-0.276, -0.041]$, Holm $p = 0.013$. The effect is economically modest. Trailing volatility alone has a

TABLE V
PRIMARY HYPOTHESES (ONE-SIDED TESTS, HOLM-ADJUSTED ACROSS H1–H3; $N = 260$ FIRMS WITH A FULL-WINDOW RETURN).

	Test	Estimate	95% CI	p	p_{Holm}
H1	IC of balanced composite with return	−0.094	[−0.219, 0.036]	0.935	1.000
H2	Partial ρ(ESG, realised vol. | trailing vol.)	−0.163	[−0.276, −0.041]	0.004	0.013
H3	Top-19 minus universe return (pp)	1.9	[−10.9, 11.5]	0.543	1.000

rank correlation of 0.82 with realised volatility, so most of the raw ESG–volatility association ($\rho = -0.30$) is past volatility carried forward.

Secondary evidence. Figure 2 and Table VI give ICs for all factors and composites. No composite has a positive IC at any horizon. The only Holm-significant full-window IC is negative, for the S pillar (−0.207). At one month, the financial and market scores have significantly *negative* ICs, consistent with short-term reversal. Table VIII decomposes the risk effect:

- In rank regressions with trailing volatility, trailing beta, size, and sector and country fixed effects, the composite (−0.085, $p = 0.019$) and the S pillar (−0.103, $p = 0.014$) predict lower realised volatility.
- The G pillar has a pooled partial correlation of −0.238 with realised *beta* but none with volatility (0.005, $p = 0.908$). The beta association is a pooling effect: it disappears once sector and country fixed effects are added (Table IX).
- The E pillar predicts neither.
- The part of ESG that is orthogonal to the other factors predicts neither at the 5% level.

The forward risk signal is therefore carried mostly by the financial-proxy S pillar. Section V-D asks what that signal is.

The cost of the tilt. Table VII shows what the risk tilt cost in a rising market. The top-19 ESG-only portfolio's excess return over the universe was −19.9 pp (CI $[-27.9, -10.2]$). Adjusting for trailing beta leaves −17.4 pp. Sector-and-country neutralisation reduces the gap to −8.4 pp (CI $[-16.2, 0.5]$): roughly half of the shortfall is a sector bet. The portfolio overweights utilities and energy and underweights technology, which was the best-performing sector.

D. RQ4: Where the Risk Signal Comes From

H2 is the only positive result, so we test it hardest. Figure 3 summarises the evidence, and Tables IX and X give the details. The answer has three parts.

It survives known risk factors. With full controls (trailing volatility and beta, size, quality, leverage, liquidity, momentum, value, and sector and country effects), the ESG coefficient is −0.106 (95% CI $[-0.190, -0.022]$, one-sided $p = 0.006$; H4). Two further checks agree.

- *Double sort.* Within trailing-volatility terciles, high-ESG firms had −6.1 pp lower realised volatility than low-ESG firms (CI $[-9.1, -3.3]$). The gap grows from −2.8 pp in the calmest tercile to −11.7 pp in the most volatile.
- *Robustness.* The effect holds without the 2 takeover-pinned firms, with winsorised volatility, without utilities and real estate, and has the same sign in both countries (US −0.208, India −0.267).

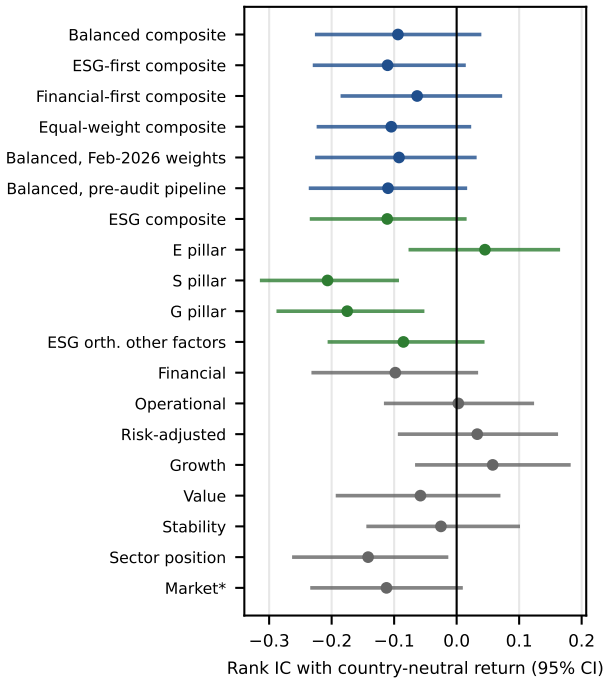


Fig. 2. Out-of-sample rank ICs over (2026-04-02, 2026-09-23] with 95% bootstrap intervals. Blue: composites; green: ESG measures; grey: financial factors (*market score carries zero weight). Two-sided 80%-power detection threshold: $|\rho| \geq 0.173$.

TABLE VI
OUT-OF-SAMPLE RANK ICS WITH COUNTRY-NEUTRAL RETURNS. BOLD: HOLM $p < 0.05$ WITHIN HORIZON.

Signal	1m	3m	Full	95% CI (full)
ESG composite	−0.138	−0.078	−0.111	[−0.235, 0.016]
E pillar	−0.104	0.018	0.045	[−0.077, 0.166]
S pillar	−0.116	−0.164	**−0.207**	[−0.315, −0.092]
G pillar	−0.025	−0.118	−0.175	[−0.288, −0.052]
ESG ⊥ other factors	−0.054	−0.077	−0.085	[−0.206, 0.045]
Financial	**−0.190**	−0.099	−0.098	[−0.232, 0.034]
Operational	−0.042	0.051	0.003	[−0.116, 0.124]
Risk-adjusted	−0.038	0.096	0.033	[−0.094, 0.162]
Growth	0.032	0.038	0.058	[−0.067, 0.182]
Value	0.021	−0.120	−0.058	[−0.193, 0.070]
Stability	−0.154	−0.076	−0.025	[−0.144, 0.101]
Market (not weighted)	**−0.208**	−0.073	−0.112	[−0.234, 0.010]
Sector position	−0.098	−0.074	−0.142	[−0.263, −0.013]
Balanced composite	−0.175	−0.074	−0.094	[−0.227, 0.040]
ESG-first composite	−0.159	−0.082	−0.111	[−0.230, 0.015]
Financial-first composite	−0.168	−0.051	−0.063	[−0.186, 0.073]
Equal-weight composite	−0.160	−0.091	−0.105	[−0.224, 0.023]
Balanced, Feb-2026 weights	−0.171	−0.078	−0.092	[−0.226, 0.032]
Balanced, pre-audit pipeline	**−0.189**	−0.091	−0.110	[−0.237, 0.017]

The ESG–beta association does not survive: it falls from −0.138 ($p = 0.003$) with the trailing control to −0.033 ($p = 0.562$) with full controls. No ESG measure predicts drawdowns.

It is not in the measured ESG data. The signal sits in

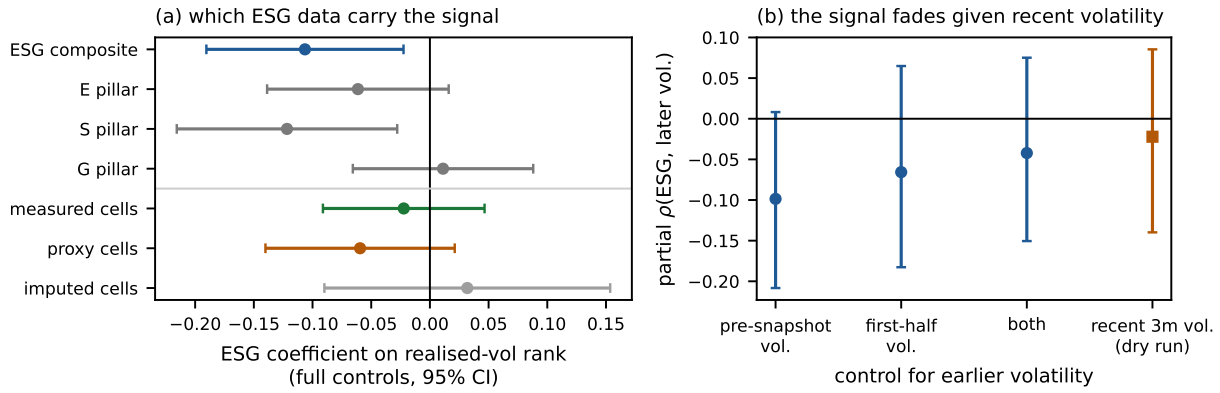


Fig. 3. Where the ESG risk signal comes from. (a) Coefficient of each ESG measure in a rank regression of realised OOS volatility with all 8 controls and sector and country fixed effects. (b) Partial Spearman correlation of ESG with realised volatility after 2026-06-28, given earlier volatility. The orange point is the registered dry run: volatility over (2026-07-01, 2026-09-23] given the preceding months. 95% intervals.

TABLE VII
TOP-19 PORTFOLIO EXCESS RETURN OVER THE EQUAL-WEIGHTED UNIVERSE (PP, COUNTRY-NEUTRAL). β-ADJ.: MARKET-MODEL ADJUSTMENT WITH TRAILING BETA; SECT.-NEUT.: SECTOR-AND-COUNTRY-NEUTRAL FULL-WINDOW RETURNS.

Top-N by	1m	3m	Full	β-adj.	Sect.-neut.	95% CI (full)
Balanced composite	2.2	0.9	1.9	1.6	3.1	[−11.2, 11.0]
ESG-first composite	−0.6	−5.7	−9.4	−8.6	−0.8	[−18.5, 0.8]
Financial-first composite	1.5	2.5	0.9	0.3	−1.3	[−10.5, 13.8]
Equal-weight composite	0.3	−0.9	−2.8	−3.8	1.0	[−11.4, 7.7]
Balanced, Feb-2026 weights	0.5	−1.0	2.1	1.3	6.4	[−9.7, 16.2]
Balanced, pre-audit pipeline	1.2	−2.3	−3.4	−3.2	−2.8	[−12.6, 4.9]
ESG composite	−1.1	−8.0	−19.9	−17.4	−8.4	[−27.9, −10.2]

TABLE VIII
ESG AND REALISED RISK. PARTIAL ρ CONTROLS FOR THE TRAILING VALUE OF THE SAME RISK MEASURE; BOLD: HOLM $p < 0.05$ ACROSS THE 15 SECONDARY TESTS. REGRESSION: RANK OF REALISED VOLATILITY ON ESG RANK WITH TRAILING VOLATILITY, BETA, SIZE, AND SECTOR AND COUNTRY EFFECTS (HC3).

	Realised volatility		Realised β	Vol. regression	
ESG measure	raw ρ	partial ρ	partial ρ	coef.	p
ESG composite	−0.303	−0.163	**−0.239**	−0.085	0.019
E pillar	−0.151	−0.071	−0.067	−0.052	0.142
S pillar	−0.285	−0.123	**−0.330**	−0.103	0.014
G pillar	−0.252	−0.125	**−0.238**	0.005	0.908
ESG $\perp$ other factors	−0.184	−0.154	−0.071	−0.056	0.093

the S pillar (-0.122, $p = 0.011$), whose inputs are almost all financial proxies. The G pillar (0.011) carries none of it. Splitting each firm's cells by source makes this sharper:

- *Measured cells* (ISS, SEC, registry) have partial $\rho = 0.032$ with realised volatility (CI $[-0.10, 0.16]$), and are nearly independent of the financial factors ($R^2 = 0.07$).
- *Proxy cells* are largely financial ($R^2 = 0.57$) and have partial $\rho = -0.086$ (CI $[-0.20, 0.04]$).
- *Imputed cells* have a strong raw association (-0.269), which is a sector effect: it vanishes once sector fixed effects are included (Table IX).

TABLE IX
RANK-REGRESSION COEFFICIENTS OF EACH ESG MEASURE FOR REALISED RISK. "TRAILING": CONTROL FOR THE PRE-SNAPSHOT VALUE OF THE SAME RISK MEASURE; "FULL": ALL CONTROLS OF SECTION IV-C, WITH SECTOR AND COUNTRY FIXED EFFECTS (HC3). DRAWDOWN IS SIGNED, SO A POSITIVE VALUE MEANS A SHALLOWER DRAWDOWN. BOLD: $p < 0.05$ (TWO-SIDED, UNADJUSTED). $N = 260$.

	Realised vol.		Realised β		Drawdown
ESG measure	trailing	full	trailing	full	full
ESG composite	**−0.087**	**−0.106**	**−0.138**	−0.033	0.060
E pillar	−0.048	−0.061	**−0.103**	0.000	0.063
S pillar	**−0.103**	**−0.122**	**−0.114**	0.035	0.089
G pillar	−0.001	0.011	−0.055	−0.043	−0.088
Measured-cell ESG	−0.018	−0.022	−0.064	−0.021	−0.069
Proxy-cell ESG	−0.067	−0.059	**−0.144**	0.048	0.071
Imputed-cell ESG	0.025	0.032	0.047	−0.056	0.094

The two non-imputed sub-scores are almost unrelated ($\rho = 0.15$). Adding measured-cell ESG to the out-of-fold volatility forecast changes R^2 by -0.0021.

It is small, and partly stale volatility. Adding the composite to a cross-validated volatility forecast raises out-of-fold R^2 from 0.729 to 0.734, an improvement in 100% of repeats but only 0.0056 in size. Recent volatility erodes it further. Given pre-snapshot volatility, ESG's partial correlation with second-half volatility is -0.099. Given first-half realised volatility it is -0.066 (CI $[-0.18, 0.06]$, $p = 0.292$), and given both it is -0.042. The monthly partial correlations are negative in 6 of 6 months, but each is small (-0.111 to -0.035). ESG therefore behaves like a slow-moving summary of a firm's risk profile. It adds a little beyond long-horizon trailing volatility and little beyond recent volatility.

E. Robustness of the Return Null

The failure of H1 is not a consequence of the chosen weights (Figure 4).

TABLE X

ROBUSTNESS OF THE ESG–VOLATILITY RESULT. UPPER PANEL: H2 AND H4 ON SUBSAMPLES. LOWER PANEL: PARTIAL SPEARMAN CORRELATION OF ESG WITH REALISED VOLATILITY AFTER 2026-06-28, GIVEN EARLIER VOLATILITY.

Sample / test	N	H2 partial ρ [95% CI]	H4 coef. [95% CI]
all (pre-declared)	260	-0.163 [-0.27, -0.04]	-0.106 [-0.19, -0.02]
excl. deal-pinned	258	-0.166 [-0.27, -0.04]	-0.104 [-0.18, -0.03]
winsorised vol 1/99	260	-0.163 [-0.27, -0.05]	-0.106 [-0.19, -0.02]
US only	172	-0.208 [-0.35, -0.06]	-0.067 [-0.17, 0.04]
India only	88	-0.267 [-0.46, -0.04]	-0.174 [-0.41, 0.06]
excl. Utilities, Real Estate	233	-0.171 [-0.29, -0.04]	-0.110 [-0.20, -0.02]
2nd-half vol. | pre-snapshot vol.	260	-0.099 [-0.21, 0.01]	–
2nd-half vol. | 1st-half vol.	260	-0.066 [-0.18, 0.06]	–
2nd-half vol. | both	260	-0.042 [-0.15, 0.08]	–

- *Random weights.* Across 2000 weight vectors drawn uniformly from the simplex, the OOS IC has median -0.087 (2.5–97.5% range $[-0.141, 0.008]$) and is positive in only 3.3% of draws. Around the deployed weights ($\kappa = 100$), it is positive in 0.0% of draws.
- *ESG weight.* More ESG weight lowers the IC (rank correlation -0.30 across draws). The IC moves from -0.055 with no ESG to -0.111 with ESG only.
- *Low volatility without ESG.* The top-19 portfolio has lower realised volatility than the universe in 98% of draws, whatever the ESG weight (rank correlation -0.05). With no ESG weight, its volatility is already -7.0 pp below the universe. The composite's low-risk profile comes from its risk-adjusted and stability factors, not from ESG.
- *Over time.* Monthly ICs of the balanced composite are negative in 3 of 6 months (range -0.175 to 0.060).
- *Costs.* Round-trip costs for the top-19 portfolio are 0.21 pp, which leaves a net excess of 1.7 pp (1.3 pp at three times the costs). The sign of H3 is thus unaffected, and it remains insignificant.

F. Rank Uncertainty

Figure 5 and Table XI report rank intervals for the balanced profile. For the top-19 names, the median 95% interval spans 15 ranks (maximum 28). Weight uncertainty dominates: alone it gives 13 ranks, against 3 for universe resampling. 18 of the 19 current constituents remain in the top-19 in at least half of the draws, and the expected overlap between the published and a perturbed portfolio is 16.5 names. Rankings near the top are therefore reasonably determined *given* the construction; the OOS results show that the construction itself does not predict returns.

VI. DISCUSSION

A. Public-Data ESG Is Mostly Financial Quality

The construct-validity audit has a simple explanation. When direct disclosure is missing, each proxy is chosen for a plausible link between a financial ratio and an ESG outcome: margins and emissions [6], revenue per employee and satisfaction [7], cash returned and community investment. Each link is defensible in isolation. Aggregated, however, they turn the E and S pillars into a within-sector profitability and payout score. That score is largely spanned by the financial factors, and it carries the index's whole forward risk signal. The G pillar, built on third-party governance assessments, is the only component that adds information roughly orthogonal to the rest, yet that information did not predict risk or returns in our window. The practical implications are:

1) report provenance and redundancy per pillar, not just coverage;
2) do not describe proxy E and S pillars as environmental or social performance;
3) treat governance scores as the only component that currently adds independent information.

TABLE XI

TOP-19 BALANCED PORTFOLIO: 95% RANK INTERVAL, PROBABILITY OF REMAINING IN THE TOP-19, AND REALISED FULL-WINDOW RETURN (†NSE; "EXIT": ACQUIRED DURING THE WINDOW).

Rank	Ticker	Sector	95% CI	P(top-N)	OOS ret. (%)
1	CTRA	Energy	[1, 2]	1.00	exit
2	INCY	Healthcare	[1, 2]	1.00	29.9
3	GGG	Industrials	[3, 6]	1.00	-6.7
4	DECK	Consumer Cyclical	[3, 11]	1.00	-20.1
5	STLD	Basic Materials	[3, 10]	1.00	30.3
6	NAVINFLUOR†	Basic Materials	[3, 8]	1.00	43.7
7	STAG	Real Estate	[4, 8]	1.00	1.8
8	NNN	Real Estate	[6, 27]	0.93	-0.7
9	EWBC	Financial Services	[6, 21]	0.96	15.7
10	PIDILITIND†	Basic Materials	[6, 16]	1.00	25.6
11	MUTHOOTFIN†	Financial Services	[7, 19]	0.99	-9.4
12	MTCH	Communication Services	[9, 30]	0.68	30.3
13	CDNS	Technology	[9, 25]	0.87	10.9
14	IBKR	Financial Services	[8, 31]	0.76	35.4
15	ITT	Industrials	[10, 25]	0.84	10.1
16	VEEV	Healthcare	[9, 31]	0.73	55.6
17	RPM	Basic Materials	[11, 29]	0.52	3.2
18	INVH	Real Estate	[11, 39]	0.42	6.5
19	ATO	Utilities	[9, 34]	0.60	-16.4

B. Why In-Sample Evidence Misled

The four artefacts in Section V-B share a structure: the validation target was not independent of the thing being validated.

- Trailing returns overlap the price history used by the risk-adjusted and stability factors.
- The "forward" quality proxy shares inputs with the factors, and the isotonic correction was fitted to it.
- The bootstrap perturbed scores instead of data, so it measured the injected noise.

Every such design yields favourable numbers whether or not the index works. This is the mechanism behind the replication failures documented for factor research [14], [15]. The defence is procedural: freeze the construction at a date, and test on outcomes that occur after that date.

C. ESG as a Risk Tilt, and Its Price

The forward evidence supports a risk channel, but a narrow one. Higher ESG predicts lower volatility beyond trailing volatility and 8 other characteristics, in line with [12]. It does not predict beta or drawdowns once sector and country are

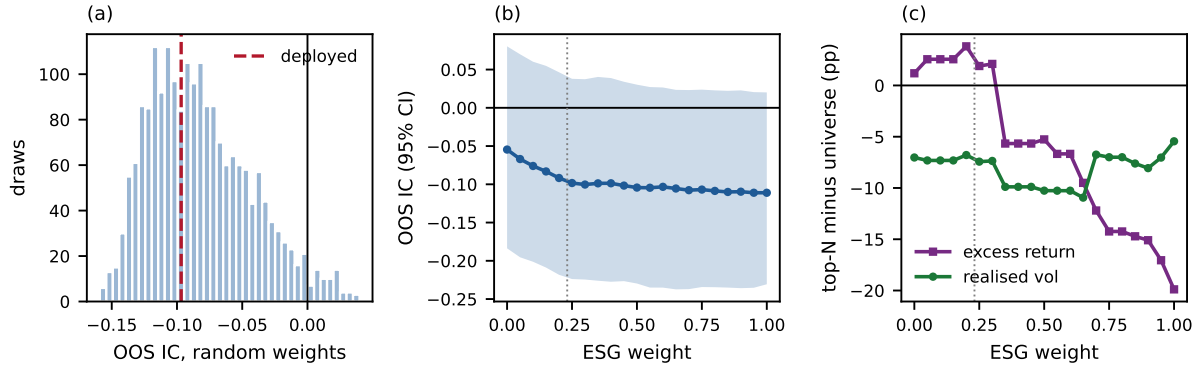


Fig. 4. Weight uncertainty and the ESG weight. (a) OOS IC of the composite under 2000 weight vectors drawn uniformly from the simplex; dashed line: deployed weights. (b) OOS IC as the ESG weight rises from 0 to 1, with the other weights in their deployed proportions (95% bootstrap band). (c) The top-19 portfolio's excess return and realised volatility relative to the universe on the same sweep. Dotted line: deployed ESG weight.

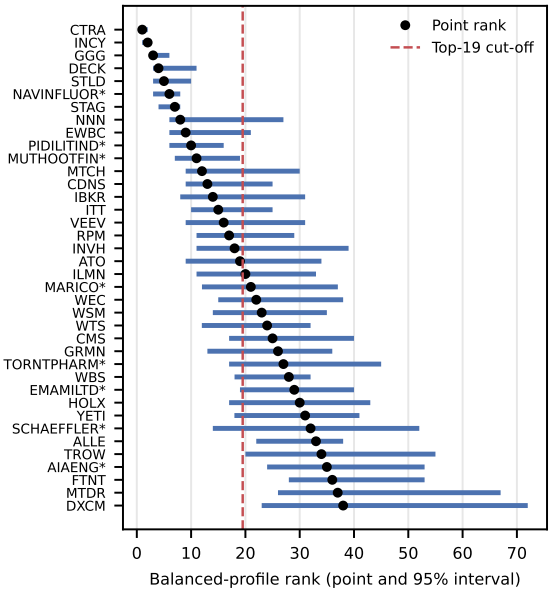


Fig. 5. Balanced-profile ranks with 95% intervals combining universe resampling and Dirichlet ($\kappa = 100$) weight perturbation ($B = 1000$). *NSE-listed.

controlled. The decomposition in Section V-D limits what this channel can be credited to:

- the measured ESG cells carry none of the signal;
- the proxy-heavy S pillar carries all of it;
- recent realised volatility absorbs most of it.

The defensible reading is that public-data ESG, as built here, is a slow-moving low-risk characteristic assembled from financial data. It is not evidence that environmental or social performance lowers risk. The evidence also does not support a return channel over six months. In a window in which both markets rose, the risk tilt was expensive, because an ESG-only screen loaded on defensive sectors and avoided technology. This is the trade-off that equilibrium models predict: sustainable assets can serve as hedges while offering lower expected returns [20], [21]. A single six-month window cannot estimate a premium. It can, however, show the size of the regime risk that an ESG-tilted mid-cap portfolio takes on relative to its universe.

D. Implications for Index Builders

Five recommendations follow from the evidence.

1) *Provenance per cell and redundancy per pillar* should be standard disclosures for any ESG index built from public data.
2) *Freeze and date* the construction, and treat any number computed on the snapshot cross-section as descriptive.
3) *Report OOS results with power.* Report the minimum detectable effect: with a few hundred firms, a single cross-section can reject only large effects.
4) *Attribute signals to sources.* Before an out-of-sample effect is credited to ESG, re-estimate it separately on measured, proxy and imputed cells, and control for recent risk.
5) *Generate the paper from the code.* In this project, every discrepancy between the manuscript and the pipeline found by the audit was a hand-copied or hand-written number.

VII. Limitations and Threats to Validity

1) **One window, one cross-section.** The OOS test covers 118 trading days in a rising market. With $N = 260$, only $|\rho| \geq 0.173$ is detectable at 80% power. H1 and H3 therefore exclude large effects, not small ones, and the risk result (H2) may not generalise to other regimes. The registered second window adds a second, independent period; repeating the snapshot–hold-out cycle beyond it is the most direct way to accumulate power.
2) **Hypotheses fixed after the window ended.** The hypotheses and analysis code were fixed before OOS prices were downloaded, but after the calendar window had passed, and they were not externally registered. Two

further facts bear on this. First, the profile weights were revised in July 2026, inside the window. Second, the authors' general knowledge of market conditions could not be excluded. The safeguards are the frozen snapshot artefacts and full reporting of alternative weightings (the February 2026 weights, and 96.7% of random weightings, give the same return conclusion). The mechanism analyses of Section V-D were specified after H1–H3 had been evaluated and are exploratory. The second window (Section IV-D) is registered before it opens, so that H2 and H4 can be confirmed or refuted there.

3) **Proxy ESG.** Our E and S pillars are not measurements of environmental or social performance (Section V-A). The findings concern public-data ESG as constructed here. They do not necessarily transfer to vendor ratings, which have their own validity problems [2], [5].
4) **Governance data dependence.** The governance pillar relies on ISS scores redistributed by a free data interface. They are available for 209 of 269 firms and are sector-imputed otherwise. Their continued availability is not guaranteed.
5) **Universe construction.** The candidate list was assembled from index constituents at the time of collection. It is not a point-in-time membership history, so earlier-period survivorship cannot be excluded. The investability filter and the OOS survivorship bounds address only the evaluation window.
6) **Multi-market pooling.** US and Indian firms are scored against a common reference, including 45 large caps. Country-neutral returns remove market-level differences in outcomes but not differences in disclosure regimes. The ESG–volatility association has the same sign in both country subsamples (Table X), but at $N = 88$ the Indian estimates are imprecise.
7) **Transaction costs and capacity.** Portfolio results are equal-weighted and single-period. We deduct stylised round-trip costs (Section V-E), but do not model market impact at scale.
8) **Provenance split.** The measured/proxy/imputed subscores average indicator ranks with equal weights, not with the pillar and materiality weights of the composite. They show where the signal lives, not how the composite would behave with only one data source. The measured cells are mostly governance items, so a null for measured cells is not a test of measured environmental data. That test requires emissions and workforce disclosures (e.g. BRSR for the Indian firms), which we leave to future work.

VIII. Conclusion

We built a transparent nine-factor ESG–financial index for 269 US and Indian mid-caps and audited it end to end. The audit answers four questions.

1) *What does public-data ESG measure?* Its E and S pillars are mostly financial quality measured within sector (other factors explain 0.42 and 0.51 of their variance). Only the governance pillar carries largely independent information.
2) *Is common in-sample evidence informative?* No. Trailing-return "ICs", a contemporaneous "forward" proxy, a correction fitted to that proxy and a misapplied bootstrap together made a composite with no demonstrable predictive power look strong.
3) *What happens out of sample?* The composite does not predict returns (IC $= -0.094$), and its top-19 portfolio does not beat the universe. Higher ESG does predict lower future volatility beyond trailing volatility (partial $\rho = -0.163$, Holm $p = 0.013$). In a rising market this risk tilt was costly: an ESG-only top-19 screen had an excess return of -19.9 pp.
4) *Is that risk signal ESG?* Only in name. It survives 8 known risk characteristics and sector effects (-0.106), but it comes from financial-proxy cells (measured cells: 0.032), adds 0.0056 to an out-of-fold volatility forecast, and falls to -0.066 given recent realised volatility.

The contribution is as much methodological as empirical. Transparency is necessary but not sufficient for validity. Provenance per cell, redundancy per pillar, a dated construction and an out-of-sample test with fixed hypotheses are cheap to implement and should accompany any ESG index built from public data.

Future work. Future work should:

- evaluate the pre-registered window (2026-09-28 to 2027-03-31), which will confirm or refute H2 and H4 against recent-volatility controls;
- repeat the snapshot–hold-out cycle quarterly, to accumulate power;
- replace financial proxies with measured E and S data (EPA TRI, BRSR filings for Indian firms, and text-mined sustainability reports [29]);
- test whether the ESG–volatility association persists across market regimes, and whether measured E and S data share it.

Reproducibility. The code, configuration, cached raw data, frozen pre-audit outputs and the registration package are in the project repository (https://github.com/Shashwat1729/esg-hybrid-index).

- `run_all.py --skip-download` rebuilds every table, figure and number in this paper from the 2026-04-02 snapshot and the cached post-snapshot prices.
- Step 25 with `--refresh` re-downloads the prices.
- Step 28 (`evaluate`) runs the registered test once the second window has closed.

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APPENDIX A
ESG INDICATORS AS ACTUALLY CONSTRUCTED

Table XII lists every ESG indicator that carries cross-sectional variation in the mid-cap universe, together with the source that actually fills it. "Sector pct." means the within-sector percentile of the stated ratio, clipped to $[2, 98]$. When a source is missing, a cell falls through to the next tier of Section III-B. The other 13 declared indicators have no source for any mid-cap firm and are excluded from scoring. They are: Scope 2 and 3 emissions, emissions intensity, energy efficiency, water intensity, environmental fines, R&D intensity (E innovation), employee turnover, women in management,

pay gap, injury rate, CEO pay ratio and the anti-corruption policy flag.

APPENDIX B
SASB MATERIALITY WEIGHTS

APPENDIX C
AUDIT LOG

All corrections below were made before any post-snapshot price was downloaded.

1) *ESG direction.* `scope1_emissions` and `esg_risk_rating` were removed from the lower-is-better set, because both are stored as higher-is-better scores.
2) *Imputation noise.* Multiplicative $\mathcal{N}(1, 0.04)$ noise in sector and global median imputation was removed.
3) *Registry data.* Registry memberships are encoded as 0/1 and no longer overwrite continuous emissions and renewable-share columns. A mapping of a 40–70 score onto the binary anti-corruption flag, which made that flag constant, was removed.
4) *Source precedence.* ISS governance scores now take precedence over SEC financial items mapped onto governance columns. A syntax error in this step had prevented the download script from running.
5) *Monotonicity correction.* The isotonic fit to the validation proxy is disabled by default (`monotonicity_correction: false`).
6) *Preference transforms.* The undocumented bottom-30% quality gate and the power exponent of 1.5 are disabled by default.
7) *Snapshot consistency.* FX is taken at the snapshot rate. `market_share` is recomputed after currency conversion. Firms that did not trade on the snapshot date are removed.
8) *Portfolio size.* $N^\star = \lceil 0.07N \rceil$, bounded to $[10, 50]$, replaces a rounding rule with a special case for $N = 276$.
9) *Rank uncertainty.* The score-perturbing "Bayesian bootstrap" was replaced by the procedure of Section IV-E.
10) *Manuscript.* Several results in an earlier draft could not be reproduced from the code and were removed. These were walk-forward ICs, Fama–MacBeth estimates, an ablation table, a claimed sign flip of the value factor, and "100% of periods" out-performance of a momentum proxy. Two further claims were withdrawn:
    - a large-cap "generalisation" statistic ($\rho \approx 1$), which measured only the effect of adding 45 firms to the percentile reference set;
    - a Kendall's W that measured agreement between investor profiles, not generalisation.

    A residualisation of E on the financial score, described in the draft, was never executed by the pipeline and is not claimed here.

TABLE XII
SOURCE OF EACH INFORMATIVE ESG INDICATOR (MID-CAP UNIVERSE).

Pillar	Indicator (column)	Actual construction
E	scope1_emissions	Sector pct. of operating margin (higher margin $\Rightarrow$ assumed lower emissions intensity) [6]
E	renewable_energy_pct	Sector pct. of revenue / market capitalisation
E	waste_recycling_pct	Sector pct. of gross margin [27]
E	carbon_reduction_target	1 if listed as CDP reporter, SBTi participant or RE100 member, else 0
S	gender_diversity_pct	Sector pct. of R&D / revenue (operating-margin fallback)
S	safety_training_hours	Mean of sector pct. of log employees and of operating margin
S	employee_satisfaction	Sector pct. of revenue per employee [7]
S	revenue_per_employee	Revenue per employee (direct ratio; duplicates the previous row's input)
S	community_investment_pct	Sector pct. of dividend yield + FCF / revenue
S	supply_chain_audit_pct	Sector pct. of payout ratio + current ratio [47]
S	human_rights_policy	1 if UN Global Compact signatory, else 0
G	board_independence_pct	ISS board risk, $(10 - r) \times 10$; else SEC dividend item; else sector median
G	exec_comp_esg_linked	ISS compensation risk; else SEC share-based compensation item
G	shareholder_rights_score	ISS shareholder-rights risk; else sector pct. of $1/(\text{D/E})$ + dividend yield
G	ethics_compliance_score	ISS audit risk; else sector median
G	esg_risk_rating	ISS overall governance risk; else sector median
G	tax_transparency_score	SEC effective tax rate (banded); else institutional ownership (%)
G	data_privacy_score	Insider ownership, banded (peak score at 10–30%)
G	esg_controversy_score	Analyst coverage, $20 + 2.5\times$ count, capped at 90
G	board_diversity_pct	Sector pct. of log market cap and inverse beta [48]
G	board_size	SEC operating-segment count, banded (8 firms); else sector or global median

TABLE XIII
SECTOR-SPECIFIC E/S/G PILLAR WEIGHTS (CONFIG/SASB_MATERIALITY.YAML), INFORMED BY THE SASB MATERIALITY MAP [10].

Sector	E	S	G
Technology	0.20	0.45	0.35
Healthcare	0.25	0.40	0.35
Financial Services	0.15	0.35	0.50
Energy	0.55	0.20	0.25
Industrials	0.40	0.30	0.30
Consumer Cyclical	0.25	0.40	0.35
Consumer Defensive	0.30	0.35	0.35
Basic Materials	0.50	0.25	0.25
Utilities	0.50	0.20	0.30
Real Estate	0.40	0.25	0.35
Communication Services	0.20	0.45	0.35